

Semi-skew polarity operators do have fixed sets: the exact two-dimensional classification

New counterexample for Remark 10.2 of arXiv:1708.09741

13 August 2026

Status. Full negative answer to the precise semi-skew nonexistence question. For every two-dimensional semi-skew operator with unequal weights, both coordinate axes solve the paper's fixed-point equation. In fact, these are the only nonempty solutions. They are unbounded and contain the origin on their boundary, exactly outside the class covered by the paper's nonexistence theorem.

Source question

For a nonempty $S \subseteq X$, Reem and Reich [1] use the polar

$$S^\circ = \{z \in X : \langle z, s \rangle \leq 1 \text{ for every } s \in S\}$$

and study

$$C = (GC)^\circ. \tag{1}$$

Their Proposition 8.5 excludes solutions that are bounded and contain 0 in their interior when G is semi-skew. Remark 10.2 asks whether one can prove or disprove that a semi-skew G has no nonempty solution at all.

Remark 10.2. It would be of interest to complete the classification of the set of solutions to (1.1) in the case where G is not positive definite. At the moment this task seems to be out of reach when $\dim(X) \geq 2$, as can be seen by considering Sections 6, 7 and 8. In this connection, it would be of interest to consider the case where G is semi-skew and to prove, or disprove, that (1.1) cannot have solutions $\emptyset \neq C \subseteq X$ in this case (the approach that we use in Section 8 is heavily based

Source evidence: arXiv:1708.09741, p. 20, Remark 10.2.

Proof intuition

The counterexample itself is one line: a semi-skew matrix swaps the two coordinate axes, and the polar of a full line is its orthogonal line. Thus each axis returns to itself after applying G and then polarizing.

There is also a rigid classification behind this observation. Any solution of (1) is automatically closed, convex, and contains zero. Bipolarity and the linear transformation rule for polars show that it must be

invariant under $G^{-1}G^T$. For unequal semi-skew weights this is a hyperbolic diagonal map, expanding one coordinate while contracting the other and reversing both signs. Iteration, rescaling toward zero, and closedness force both coordinate projections of every point back into the set. Convexity then leaves only the four coordinate subspaces; direct substitution retains exactly the two axes.

Theorem 1 (exact semi-skew classification in $\mathbb{R}^2$). *Let $\alpha_1, \alpha_2 \in \mathbb{R}$ have the same sign and satisfy $\alpha_1 \neq \alpha_2$. Define*

$$G = \begin{pmatrix} 0 & \alpha_2 \\ -\alpha_1 & 0 \end{pmatrix}, \quad G(x_1, x_2) = (\alpha_2 x_2, -\alpha_1 x_1).$$

This is the two-dimensional semi-skew operator of Definition 8.1 in [1]. Among all nonempty subsets $C \subseteq \mathbb{R}^2$, the solutions of $C = (GC)^\circ$ are exactly

$$L_1 = \mathbb{R} \times \{0\} \quad \text{and} \quad L_2 = \{0\} \times \mathbb{R}.$$

In particular, the nonexistence statement proposed for consideration in Remark 10.2 is false.

Proof. First, both displayed lines are solutions. Indeed,

$$GL_1 = L_2, \quad GL_2 = L_1.$$

For every linear subspace $L \subseteq \mathbb{R}^2$, its polar in the source's sense is $L^\perp$: if z is not orthogonal to some $\ell \in L$, scaling ℓ through both signs violates $\langle z, \ell \rangle \leq 1$, while orthogonal vectors give inner product zero. Hence

$$(GL_1)^\circ = L_2^\perp = L_1, \quad (GL_2)^\circ = L_1^\perp = L_2.$$

Taking, for example, $(\alpha_1, \alpha_2) = (1, 2)$ already disproves the proposed universal nonexistence.

We prove that there are no other solutions. Suppose that a nonempty $C \subseteq \mathbb{R}^2$ satisfies (1). As a polar, C is closed and convex and contains zero. Therefore GC has the same three properties, and the bipolar theorem gives

$$C^\circ = ((GC)^\circ)^\circ = GC. \tag{2}$$

On the other hand, the elementary transformation rule $(GC)^\circ = (G^T)^{-1}C^\circ$, combined with (1), gives

$$C^\circ = G^T C. \tag{3}$$

Equations (2)–(3) imply

$$C = HC, \quad H := G^{-1}G^T = \begin{pmatrix} -r & 0 \\ 0 & -r^{-1} \end{pmatrix}, \quad r := \frac{\alpha_2}{\alpha_1} > 0, \quad r \neq 1.$$

Replacing H by H^{-1} , which also leaves C invariant, allows us to assume $r > 1$.

Fix $x = (x_1, x_2) \in C$. Since $H^{2n}x \in C$, convexity and $0 \in C$ give

$$r^{-2n}H^{2n}x = (x_1, r^{-4n}x_2) \in C.$$

Closedness and $n \rightarrow \infty$ yield $(x_1, 0) \in C$. Applying the same argument to H^{-2n} yields $(0, x_2) \in C$.

If C contains a nonzero point of L_1 , invariance under all integer powers of H , followed by contraction toward zero inside the convex set, shows that it contains every positive and negative multiple of that point; hence $L_1 \subseteq C$. The same statement holds for L_2 . Thus, according to which coordinate projections occur, C is one of

$$\{0\}, \quad L_1, \quad L_2, \quad \mathbb{R}^2.$$

(If both projections occur, convexity combines the two full axes to give all of $\mathbb{R}^2$.) Finally, $\{0\}$ and $\mathbb{R}^2$ are interchanged by polarity and so do not solve (1), whereas L_1, L_2 do. $\square$

Verification and literature status

The bundled exact SymPy script checks the formula for $G^{-1}G^T$, the axis swaps, the polar-subspace calculations, the failure of the zero and full subspaces, and representative positive and negative unequal weights. These checks are independent sanity checks; the proof above is symbolic.

The result answers only the precise semi-skew subquestion of Remark 10.2. The paper's much broader request to classify solutions for arbitrary non-positive-definite G remains open. Exact-title, semi-skew, coordinate-axis, DOI, and citation searches through 13 August 2026 found no later source giving this answer or the two-dimensional classification. The search was bounded rather than a systematic bibliographic review.

Reference

[1] D. Reem and S. Reich, *Fixed points of polarity type operators*, J. Math. Anal. Appl. **467** (2018), 1208–1232; arXiv:1708.09741, v4 (2019).